Shamanth S Poojary

Udupi, Karnataka | shamanth25402@gmail.com | +91 91139 05491
LinkedIn: linkedin.com/in/shamanth-poojary | Portfolio: portfolio-woad-nine-40.vercel.app

Professional Summary

Embedded Systems Engineer with hands-on experience in firmware development, device driver implementation, and embedded GUI design. Proficient in C/C++, Python, and RTOS (FreeRTOS, RTX5) with strong expertise in microcontrollers including STM32, ESP32, and Silicon Labs EFR32. Experienced in developing low-level drivers (GPIO, I2C, SPI, UART), board bring-up, and hardware validation.

Education

N.M.A.M. Institute of Technology, B.E. in ECE Oct 2020 – May 2024

- GPA: 7.1
- **Coursework:** Embedded Systems, Microcontrollers, Real-Time Systems, Digital Electronics, Analog Electronics, Communication Systems, Internet of Things (IoT), Digital Signal Processing (DSP)

Professional Experience

Firmware Engineer, MIVI June 2026 – Present

- Developing and debugging TWS earbuds firmware — working on an ARM Cortex-M chip running RTX5 RTOS, bringing up and validating Classic Bluetooth and BLE functionality from bare-metal boot through full stack initialisation.

Associate Software Engineer, SpanIdea Systems Pvt. Ltd. June 2025 – May 2026

- Completed hands-on training in Linux System Programming, Linux Device Driver development on Raspberry Pi and BeagleV Ahead, and Bare-metal & FreeRTOS-based firmware development on STM32, ESP32-LyraT Mini, and STM32-P107 boards.
- Built and customized embedded Linux distributions using the Yocto Project, including creation of layers, recipes, and minimal images for target hardware.
- Developed a desktop GUI application using Python (PyQt) for configuring GNSS receivers and monitoring real-time navigation data over UART serial communication.
- Implemented real-time GNSS data visualization including satellite tracking, signal strength, and positional updates.
- Designed and developed a C-based automated test framework for validation and performance evaluation of multiple GNSS receiver modules.
- Performed static code analysis using Pylint and resolved code quality issues to improve maintainability and compliance.

Embedded Systems Engineer, Zilliot Technologies, Mangalore July 2024 – June 2025

- Developed embedded firmware for ESP32-based systems including Wi-Fi provisioning, RTC synchronization, and GUI development using LVGL with FreeRTOS.
- Implemented low-level drivers and hardware abstraction layers (HAL) for STM32 microcontrollers including GPIO, I2C, SPI, UART, EEPROM, and Flash memory.
- Designed and integrated embedded GUI systems using LVGL, including touchscreen interfacing and display driver development.
- Worked on Silicon Labs EFR32 platforms implementing BLE and BLE Mesh communication stacks and Flash drivers.
- Performed board bring-up, debugging, and hardware validation using tools such as logic analyzers and oscilloscopes.
- Conducted unit testing and firmware debugging to improve system reliability and reduce defects.

Intern, SpanIdea Systems Pvt. Ltd. July 2021 – August 2021

- Analyzed intermediate-frequency (IF) signals for GNSS systems and evaluated signal acquisition techniques.
- Used MATLAB to analyze correlation properties of GPS C/A codes for positioning accuracy.
- Gained foundational knowledge in satellite communication and GNSS signal processing.

Skills

- **Programming Languages:** C (Embedded C), C++, Python, Data Structures and Algorithms (DSA)
- **Operating Systems & System Programming:** Linux, Embedded Linux, Custom Linux-Yocto (Basic), Kernel cross-compilation, System Calls & Multithreading, Inter-Process Communication (IPC), File Systems, Shell Scripting
- **Embedded Systems & Firmware Development:** Bare-Metal Programming, RTOS (FreeRTOS, RTX5), ARM Cortex-M Firmware Development, Firmware Development, Device Driver Development, Board Bring-up, BSP (Board Support Package), Peripheral Drivers (GPIO, I2C, SPI, UART, ADC, PWM), Interrupt Handling (ISR)
- **Microcontrollers & Platforms:** STM32 (ARM Cortex-M), ESP32 (ESP32-S3), Silicon Labs EFR32BG24 (Wireless SoC, BLE), Texas Instruments SimpleLink CC3200 (Wi-Fi MCU), Arduino UNO (ATmega328P), Raspberry Pi (Linux-based Systems), Raspberry Pi Pico (RP2040), 8051 Microcontroller, PIC Microcontroller, ARM Architecture
- **Protocols & Communication:** I2C, SPI, UART, USB (Basic), TCP/IP, Wi-Fi (802.11), Classic Bluetooth (SPP, RFCOMM, SDP, A2DP), BLE, BLE Mesh, TWS (True Wireless Stereo) Systems, MQTT (Basic), 4-20mA Industrial Communication
- **Development Tools & Environments:** Git (Version Control), STM32CubeIDE, PlatformIO, Arduino IDE, Simplicity Studio, VS Code, CMake, Makefiles, JIRA
- **Debugging, Testing & Validation:** GDB Debugging, JTAG/SWD Debugging, Logic Analyzer, Oscilloscope, Unit Testing, Integration Testing, Firmware Debugging, Hardware Validation, Static Code Analysis (Pylint), Code Review
- **Embedded GUI & Display Systems:** LVGL (Light and Versatile Graphics Library), GUI Development for Embedded Applications, Touchscreen Integration (GT911), Display Driver Integration, SPI/RGB Display Interfaces
- **Databases & Miscellaneous:** Basic SQL, Serial Communication Tools, GNSS Fundamentals, Performance Optimization, DMA (Direct Memory Access)

Projects

GNSS Receiver Configuration Tool & Validation Testbench (Spanidea Systems Pvt. Ltd.)
Duration: Jun 2025 – Present

- Developed a desktop GUI application for configuring GNSS receivers and monitoring real-time navigation data over UART communication.
- Implemented interactive visualization of satellite and positional data using PyQt5.
- Designed and implemented a C-based automated testbench to validate and evaluate multiple GNSS receiver modules.
- Added feature enhancements and performed maintenance updates based on validation results.
- Performed static code analysis using Pylint to improve code quality and maintainability.

Technologies: C, Python, PyQt5, UART, Pylint, VS Code, Segger Embedded Studio

Air Quality Monitoring Device (Zilliot Technologies)
Duration: Feb 2025 – Jun 2025

- Developed firmware for an ESP32-based touchscreen air quality monitoring system.
- Integrated SEN66 sensor over I2C and APC1 sensor over UART to acquire environmental parameters.
- Designed LVGL-based graphical interface using SquareLine Studio and handled touchscreen events.
- Implemented Wi-Fi connectivity for real-time RTC synchronization and device configuration.
- Processed sensor data and performed application-specific calculations for user visualization.

Technologies: ESP32, C/C++, I2C, UART, LVGL, FreeRTOS, Wi-Fi, PlatformIO

Industrial Flow Meter Device (Zilliot Technologies)

Duration: Aug 2024 – Dec 2024

- Developed firmware on STM32F1 microcontroller for liquid flow measurement system.
- Implemented low-level drivers for display, EEPROM/Flash and peripheral abstraction layers for GPIO, I2C, SPI, and UART.
- Customized display driver and created bitmap assets for measurement units and device branding.
- Performed data acquisition, processing, and real-time measurement display.

Technologies: STM32, C/C++, FreeRTOS, I2C, SPI, UART, STM32CubeIDE

CogniView: Resolution-Responsive Surveillance with Face Tracking

Duration: Oct 2023 – Apr 2024

- Developed a Raspberry Pi-based surveillance system with automatic video resolution scaling (360p–1080p) upon human detection.
- Integrated OpenCV-based human detection and face recognition for target tracking.
- Implemented adaptive resolution logic to optimize storage and processing performance.
- Deployed and validated the system on Embedded Linux (Raspbian).

Technologies: Raspberry Pi, Embedded Linux, Python, OpenCV, face_recognition

Network Simulation using NS-2 (Ping Communication Analysis)

Duration: Aug 2023 – Oct 2023

- Designed network topology and simulated packet communication using NS-2.
- Analyzed round-trip time behavior under varying link and queue configurations.
- Visualized packet flow and debugged network behavior using NAM.

Technologies: NS-2 Simulator, TCL, NAM